I

# 集采计划预测模型算法与指标说明

20 2026

2023 2026

2026 9

- 70% 70%
- SMAPE

# 1

2025 2026 0.70

| 问题 | 当前实现 |
| --- | --- |
| | 2025 |
| | |
| | 0.70 |
| | |

70%

# 2

0 1

$$E = \frac{\ln(1+n)}{\ln(1+n)}$$

$n_o$ o $n_{max}$ 1 0

| 模型或环节 | 项目暴露度的作用 |
| --- | --- |
| | 2% |
| | 5% |
| | 20 |

| 模型或环节 | 项目暴露度的作用 |
| --- | --- |
|  | 4% |

6

# 3

58%

$$S = 0.38S + 0.17S + 0.14S + 0.08I + 0.08I + 0.08S + 0.07S$$

**F1**

$$\text{Precision} = \frac{M}{N} \qquad \text{Recall} = \frac{M}{N} \qquad \text{F1} = \frac{2\text{Precision}\text{Recall}}{\text{Precision} + \text{Recall}}$$

M $N_p$ $N_a$ F1

| 指标 | 去□延续基线 | □算 |
| --- | --- | --- |
|  | 55.77% | 826 ÷ 1,481 |
|  | 52.68% | 826 ÷ 1,568 |
| F1 | 54.18% |  |

$$S = 0.50\text{F1} + 0.20S + 0.15A + 0.10A + 0.05A$$

| 组成 | 权重 | 定义 |
| --- | --- | --- |
| F1 | 50% |  |
|  | 20% |  |
|  | 15% | 1 SMAPE 0 |
|  | 10% | 180 0 |
|  | 5% |  |

$$A = \text{mean}\left[\max\left(0, 1 - \frac{|\Delta \text{days}|}{180}\right)\right] \quad A = \max\left(0, 1 - \frac{|N - N|}{N}\right)$$

## 4

$$S = 0.46P + 0.20P + 0.09P + 0.08H + 0.06H + 0.05T + 0.04T + 0.02E$$

| 证据 | 权重 | □释 |
|---|---|---|
| 2025 | 46% | |
| 2024 | 20% | |
| 2023 | 9% | |
| 2025 | 8% | |
| 2025 | 6% | |
| | 5% | |
| | 4% | |
| | 2% | |

$$N = \text{round}\left(0.60N + 0.30N + 0.10N\right)$$

1,069 1,069 60% 30% 10%
EWMA

## 5

1 2 3 4

| 参数 | 当前值 | 含义 |
|---|---|---|
| | 70 | 70 |
| | 5 | 5 |
| | 12 | 12 |

| 参数 | 当前值 | 含义 |
| --- | --- | --- |
| | 4 | 20 √20 |
| | 0 1 | 0 1 |

$$\mathrm{RF}\left(X\right)=\frac{1}{70}\sum_{k=1}^{70}\mathrm{Tree}\left(X\right)$$

70

## 6

$$r=y-\hat{y}\qquad \hat{y}=\hat{y}+0.06\mathrm{Tree}\left(X\right)$$

| 参数 | 当前值 | □用 |
| --- | --- | --- |
| | 55 | 55 |
| | 0.06 | 6% |
| | 2 | |
| | 15 | 15 |

1 0.4 +0.6 0 0.7 −0.7 0.06

## 7

$$S=0.22R\left(\mathrm{EWMA}\right)+0.18\mathrm{Trend}+0.25R\left(\mathrm{RF}\right)+0.25R\left(\mathrm{GBT}\right)+0.06R\left(T\right)+0.04E$$

R 0 1

| 分量 | 权重 | 选择原因 |
| --- | --- | --- |
| | 22% | |
| | 18% | |
| | 25% | |
| | 25% | |

| 分量 | 权重 | 选择原因 |
| --- | --- | --- |
|  | 6% |  |
|  | 4% |  |

$$\text{Trend} = \max\left(0, C + 0.25\left(C - C\right)\right)$$

25% 2025 90% 110% 50% 150%

22% 18% 25% 25% 6% 4% AI
50% 40% 10%

2024 2025 2026

# 8

2026 e N(y,e) e y ln(1+x)

| 序号 | 代码字段 | 业务定义 | 计算方法 |
| --- | --- | --- | --- |
| 1 | count_lag1 | 2025 | N(2025,e) |
| 2 | count_lag2 | 2024 | N(2024,e) |
| 3 | count_lag3 | 2023 | N(2023,e) |
| 4 | present_lag1 | 2025 | 1 0 |
| 5 | present_lag2 | 2024 | 1 0 |
| 6 | present_lag3 | 2023 | 1 0 |
| 7 | recency |  | 2025 1 2024 1/2 2023 1/3 0 |
| 8 | item_demand_lag1 | 2025 | ln(1+2025 ) |
| 9 | item_demand_lag2 | 2024 | ln(1+2024 ) |
| 10 | org_demand_lag1 | 2025 | ln(1+2025 ) |
| 11 | org_demand_lag2 | 2024 | ln(1+2024 ) |
| 12 | region_demand_lag1 | 2025 | ln(1+2025 ) |
| 13 | amount_lag1 | 2025 | ln(1+ 2025 ) 0 |
| 14 | amount_lag2 | 2024 | ln(1+ 2024 ) 0 |

| 序号 | 代码字段 | 业务定义 | 计算方法 |
| --- | --- | --- | --- |
| 15 | amount_change | | amount_lag1 amount_lag2 |
| 16 | start_month | | 2025 ÷30.5÷12 0.5 |
| 17 | duration | | 2025 0 730 ÷365 0.12 |
| 18 | transaction_lag1 | | ln(1+2025 ) |
| 19 | transaction_entity_lag1 | | ln(1+2025 ) |
| 20 | project_exposure | | ln(1+ )÷ln(1+ ) |

7 6 6 1 7 7 5 1
20

# 9

P A

$$S = \max\left(0, 1 - \frac{\left|\ln(1+P) - \ln(1+A)\right|}{\ln(5)}\right)$$

800 900

$$1 - \frac{\left|\ln(801) - \ln(901)\right|}{\ln(5)} = 0.926903 \approx 92.7\%$$

92.7% ln5 5 0 1 0

$$\text{SMAPE} = \frac{|P - A|}{(|P| + |A|)\ / 2} \qquad A = \max\left(0, 1 - \min\left(\bar{\text{SMAPE}}, 1\right)\right)$$

800 900 SMAPE 100÷850 11.76% 88.24% 92.7% 88.24%
7% 15%

| 容易引起误解的说法 | 建议表述 |
| --- | --- |
| 70% 70% | 70% |
| AI | |

| □□□□□□□□□□ | □□□□ |
| --- | --- |
| | |
| | F1 |
| 1 2 | |

2026